Supplementary Material

Enhancing Corneal Endothelial Cell Image Segmentation via Generative Data Augmentation and Transfer Learning

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4) Computational Cost Discussion:

To evaluate the practical feasibility of the proposed framework, the computational costs of model training, synthetic-image generation, and segmentation inference were estimated under the desktop environment described in Section IV-A.

For Stable Diffusion 1.5 LoRA training, the computational cost increases approximately linearly with the number of training images because each image is optimized for 40 steps. For the class-specific subsets containing 5, 50, and 500 images, the estimated training times are approximately 6 minutes, 50 minutes, and 8 hours per training run, respectively. Training images with and without guttata requires similar computational time.

For object detection, the lightweight YOLOv8n, YOLO11n, and YOLO12n models are estimated to require approximately 40 minutes for 500 epochs at a resolution of 224×224. Although YOLO12n has a parameter count similar to that of YOLO11n, its attention-related operations may introduce additional computational overhead. In comparison, RT-DETR-l is estimated to require approximately 2.5 hours because its 32 million parameters and Transformer-based detection operations impose substantially higher computational and memory requirements.

At a generation resolution of 512×512, Stable Diffusion generation with and without guttata is estimated to require approximately 10 seconds per image when using 30-50 denoising steps. The two prompt conditions have comparable execution times because they use the same denoising architecture. ControlNet-conditioned generation is estimated to require approximately 10 seconds per image because of the additional conditioning branch. Consequently, generating a complete synthetic mask-image pair is expected to require approximately 20 seconds.

Segmentation training time increases with both synthetic dataset size and model complexity. For 1,000 synthetic training pairs, the overall training cost is estimated at approximately 6 hours for U-Net, 10 hours for TransUNet, and 20 hours for SwinUNet. The substantially higher cost of SwinUNet is consistent with its larger parameter count and the computational and memory overhead of window-based self-attention.

For model-only segmentation inference at 224×224 resolution, a batch size of one, and a warmed-up GPU, U-Net is estimated to process one image in approximately 10-20 ms, corresponding to 50-100 FPS. TransUNet is estimated to require 25-50 ms per image, or 20-40 FPS, whereas SwinUNet requires approximately 60-120 ms per image, corresponding to 8-17 FPS. These estimates exclude additional preprocessing operations. Overall, U-Net provides the most suitable of the three evaluated architectures for deployment on a desktop clinical workstation in this research.